# Stage 3G Physics Note: Off-Diagonal Conformal Cartoon Translation

GL-with-AI Project

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## 1 Scope

Stage 3G extends the Stage 3F diagonal conformal-cartoon translation to a reduced metric block with nonzero $h_{xz}$ or $\gamma_{xz}$. It is a derivation and symbolic-validation note only. No C++ source terms or variables are implemented, and symbolic execution alone does not establish physical correctness of an evolution.

## 2 Metric Ansatz

The reduced physical spatial metric is

$$dl^2 = \gamma_{xx}\,dx^2 + 2\gamma_{xz}\,dx\,dz + \gamma_{zz}\,dz^2 + x^2\gamma_{ww}\,d\Omega_2^2. \tag{1}$$

The gridded directions are $x, z$. The hidden transverse directions are two equivalent $w$ directions, so

$$N_{\text{hidden}} = \text{GR\_SPACEDIM} - \text{CH\_SPACEDIM} = 4 - 2 = 2. \tag{2}$$

As before, $h_{ww}$ and $\gamma_{ww}$ are hidden Cartesian-like components; the spherical/cartoon reconstruction supplies the external factor $x^2$.

## 3 Determinants

The reduced physical block determinant is

$$D_\gamma = \gamma_{xx}\gamma_{zz} - \gamma_{xz}^2. \tag{3}$$

Thus the cartoon/Cartesian-like four-dimensional physical determinant is

$$\det\gamma_{4\text{D}} = D_\gamma\,\gamma_{ww}^2. \tag{4}$$

With

$$h_{ij} = \chi\gamma_{ij}, \qquad \gamma_{ij} = \chi^{-1}h_{ij}, \tag{5}$$

the conformal factor is

$$\chi = \left[\left(\gamma_{xx}\gamma_{zz} - \gamma_{xz}^2\right)\gamma_{ww}^2\right]^{-1/4}. \tag{6}$$

For the conformal block,

$$D_h = h_{xx}h_{zz} - h_{xz}^2, \tag{7}$$

and the conformal determinant condition is

$$\det h_{4\mathrm{D}} = D_h h_{ww}^2 = \left(h_{xx}h_{zz} - h_{xz}^2\right) h_{ww}^2 = 1. \tag{8}$$

This determinant is imposed on the reduced cartoon/Cartesian-like variables; it does not include spherical-coordinate Jacobian factors.

# 4 Inverse Metric

The inverse conformal block is

$$h^{xx} = \frac{h_{zz}}{D_h}, \qquad h^{xz} = -\frac{h_{xz}}{D_h}, \qquad h^{zz} = \frac{h_{xx}}{D_h}, \qquad h^{ww} = \frac{1}{h_{ww}}. \tag{9}$$

# 5 Tracelessness

The conformal tracelessness condition is the full four-dimensional one,

$$h^{xx}A_{xx} + 2h^{xz}A_{xz} + h^{zz}A_{zz} + 2h^{ww}A_{ww} = 0. \tag{10}$$

The $2h^{xz}A_{xz}$ term is the symmetric off-diagonal contraction. The $2h^{ww}A_{ww}$ term is the hidden-direction multiplicity. Solving for the hidden component gives

$$A_{ww} = -\frac{h_{ww}}{2}\left(h^{xx}A_{xx} + 2h^{xz}A_{xz} + h^{zz}A_{zz}\right). \tag{11}$$

# 6 Extrinsic Curvature

For GR_SPACEDIM = 4,

$$K_{ij} = \chi^{-1}\left(A_{ij} + \frac{h_{ij}K}{4}\right). \tag{12}$$

The off-diagonal component is therefore

$$K_{xz} = \chi^{-1}\left(A_{xz} + \frac{h_{xz}K}{4}\right). \tag{13}$$

The trace is

$$K = \gamma^{xx}K_{xx} + 2\gamma^{xz}K_{xz} + \gamma^{zz}K_{zz} + 2\gamma^{ww}K_{ww}. \tag{14}$$

The denominator 4 is the full physical spatial dimension, not the two gridded directions and not ordinary $3+1$ intuition.

# 7 Script Checks

The repository script `docs/derivations/offdiagonal_conformal_cartoon_sympy.py` checks the physical and conformal determinant identities, the inverse metric, the explicit normalized $\det h_{4\mathrm{D}} = 1$ guard, full four-dimensional tracelessness, reconstruction of $K_{xx}, K_{xz}, K_{zz}, K_{ww}$, and a denominator guard showing that $p = 4$ works while $p = 2$ and $p = 3$ fail under a numeric substitution.

It also checks the diagonal limit

$$h_{xz} = \gamma_{xz} = A_{xz} = K_{xz} = 0, \tag{15}$$

which recovers the Stage 3F diagonal formulas.

## 8 Companion Ricci Gate

The companion repository script `docs/derivations/offdiagonal_ricci_flat_gate_sympy.py` checks the geometric Ricci engine on flat metrics with genuine off-diagonal cross terms. The constant-shear case is

$$dl^2 = (1 + \lambda^2)\, dx^2 + 2\lambda\, dx\, dz + dz^2 + x^2 d\Omega_2^2, \tag{16}$$

corresponding to $dZ = dz + \lambda dx$. The $x$-dependent shear case is

$$dl^2 = (1 + x^2)\, dx^2 + 2x\, dx\, dz + dz^2 + x^2 d\Omega_2^2, \tag{17}$$

corresponding to $dZ = dz + x\, dx$. The script verifies that the full matrix inverse is used and that all Ricci tensor components and the Ricci scalar vanish for both metrics. This is a targeted off-diagonal Ricci regression, not a full CCZ4 RHS validation.

## 9 Limitations

This note is an algebraic translation and validation scaffold. It does not derive the full CCZ4 right-hand sides, does not address regularity near $x = 0$, and does not add or validate any C++ implementation.